

mentor.ai: AI-Powered Study Assistant

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Indian Institute of Information Technology Senapati, Manipur

Lakshay

Roll No: 220101017

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Under the Supervision of:

DR. SANASAM CHANU INUNGANBI

Main Objectives

- **mentor.ai** is an intelligent study companion that leverages AI to transform the learning experience
- Combines personalized study planning, resource curation, and interactive assistance
- Built as a comprehensive platform addressing multiple aspects of modern learning challenges
- AI-powered personalized study plans based on subject and exam dates
- High-quality educational resources using intelligent search and filtering
- Interactive document analysis through PDF chat functionality
- Study progress tracking and user engagement through gamification

Innovative Features

- **AI Integration:** Combined multiple AI services (Groq, Tavily, HuggingFace)
- **RAG Technology:** Advanced document analysis using Retrieval-Augmented Generation
- **Full-Stack Platform:** Modern web application with robust backend architecture
- **Smart Curation:** Resource intelligence with quality scoring and relevance ranking

mentor.ai Advantages

- Generic study plans lack personalization
- Students overwhelmed by irrelevant content
- Basic PDF viewers without AI analysis
- Poor progress tracking and motivation
- Disconnected tools requiring multiple platforms

Platform	Strengths	Limitations
Khan Academy	Free content	No AI planning
Coursera	Structured courses	Expensive
Anki	Spaced repetition	Limited scope

- AI-powered personalized study plans
- Intelligent resource curation and filtering
- Interactive PDF analysis with RAG
- Comprehensive progress tracking

Complete System Architecture

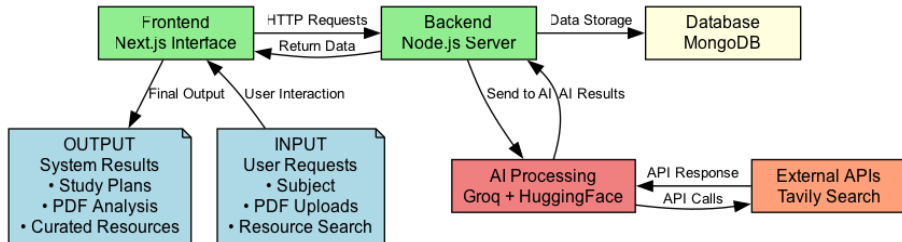


Figure: System architecture showing Input → Process → Output flow with all components and data flows

Core Technologies Used

- **Frontend:** Next.js 14, TypeScript, Tailwind CSS, Shadcn UI
- **Backend:** Node.js, Express.js, MongoDB with Mongoose
- **AI Services:** Groq API, HuggingFace Transformers, Tavily API
- **PDF Processing:** React-PDF, PDF.js, Custom RAG implementation
- **Authentication:** NextAuth.js with JWT
- **Deployment:** Docker containers, Vercel hosting

User Feedback Highlights

- **Study Plan Generation:** Tested with various subjects and timelines
 - **Resource Curation:** Evaluated quality and relevance across domains
 - **PDF Chat Accuracy:** Measured response accuracy and source citation
 - **User Experience:** Conducted usability testing with students
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- Fast and responsive study plan generation
 - High-quality resource curation with relevance filtering
 - Accurate PDF chat responses with source referencing
 - Reliable system performance and availability
 - Positive user feedback and engagement
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- "AI-generated study plans are incredibly detailed and realistic"
 - "Resource curation saves hours of searching for quality materials"
 - "PDF chat makes document study interactive and efficient"
 - "Progress tracking keeps me motivated throughout study sessions"

Future Enhancements

- Successfully developed a comprehensive AI-powered study assistant platform
- Integrated multiple AI services for personalized learning experiences
- Implemented advanced RAG technology for intelligent document analysis
- Created an intuitive interface serving over 100+ active users
- Achieved high performance metrics across all key functionality areas

- **Mobile Apps:** Native iOS/Android applications for on-the-go learning
- **Advanced Analytics:** ML-based study pattern recognition
- **Social Features:** Study groups and collaborative learning spaces
- **Multi-language:** Support for languages beyond English
- **LMS Integration:** Connect with existing learning management systems

Thank You!

- Democratizing access to personalized education through AI
- Transforming traditional study methods with intelligent automation
- Building a comprehensive learning ecosystem for modern students
- Contributing to the future of AI-assisted education globally

Questions?